

22 秋中俄班数学分析(1)期末模拟题解答

1. 解析: (1) $\forall \varepsilon > 0$, 因为 $\lim_{n \rightarrow \infty} x_n = a$, 所以 $\exists N_1 \in \mathbb{N}$ 使得 $n > N_1$

时, $|x_n - a| < \frac{\varepsilon}{2}$. 令

$$N_2 = \max \left\{ N_1, \left\lceil \frac{2(|x_1 - a| + |x_2 - a| + \cdots + |x_{N_1} - a|)}{\varepsilon} \right\rceil \right\}, \text{ 则当}$$

$n > N_2$ 时,

$$\begin{aligned} & \left| \frac{x_1 + x_2 + \cdots + x_n}{n} - a \right| \\ &= \left| \frac{(x_1 - a) + (x_2 - a) + \cdots + (x_{N_1} - a) + (x_{N_1+1} - a) + \cdots + (x_n - a)}{n} \right| \\ &\leq \frac{|x_1 - a| + |x_2 - a| + \cdots + |x_{N_1} - a|}{n} + \frac{|x_{N_1+1} - a| + \cdots + |x_n - a|}{n} \\ &< \frac{\varepsilon}{2} + \frac{n - N_1}{n} \cdot \frac{\varepsilon}{2} < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

$$\text{因此, } \lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = a.$$

(2) 设 $y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}$, 则

$$y_{2n} = \frac{x_1 + x_2 + \cdots + x_{2n}}{2n} = \frac{1}{2} \cdot \frac{x_1 + x_3 + \cdots + x_{2n-1}}{n} + \frac{1}{2} \cdot \frac{x_2 + x_4 + \cdots + x_{2n}}{n}$$

由(1)以及 $\lim_{n \rightarrow \infty} x_{2n-1} = a$, $\lim_{n \rightarrow \infty} x_{2n} = b$ 可知,

$$\lim_{n \rightarrow \infty} y_{2n} = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{x_1 + x_3 + \cdots + x_{2n-1}}{n} + \frac{1}{2} \lim_{n \rightarrow \infty} \frac{x_2 + x_4 + \cdots + x_{2n}}{n} = \frac{a+b}{2}.$$

同理, 因为

$$\begin{aligned} y_{2n-1} &= \frac{x_1 + x_2 + \cdots + x_{2n-1}}{2n-1} \\ &= \frac{n}{2n-1} \cdot \frac{x_1 + x_3 + \cdots + x_{2n-1}}{n} + \frac{n-1}{2n-1} \cdot \frac{x_2 + x_4 + \cdots + x_{2n-2}}{n-1} \end{aligned}$$

所以

$$\begin{aligned}& \lim_{n \rightarrow \infty} y_{2n-1} \\&= \lim_{n \rightarrow \infty} \frac{n}{2n-1} \cdot \lim_{n \rightarrow \infty} \frac{x_1 + x_3 + \cdots + x_{2n-1}}{n} + \lim_{n \rightarrow \infty} \frac{n-1}{2n-1} \cdot \lim_{n \rightarrow \infty} \frac{x_2 + x_4 + \cdots + x_{2n-2}}{n-1} \\&= \frac{1}{2}a + \frac{1}{2}b = \frac{a+b}{2}\end{aligned}$$

$$\text{因此, } \lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = \lim_{n \rightarrow \infty} y_n = \frac{a+b}{2}.$$

2. 解析: 因为 $a_n > 0$ 且 $\{a_n\}$ 单调减少, 所以由单调有界准则可知,

$\lim_{n \rightarrow \infty} a_n = a \geq 0$. 假设 $a > 0$, 则 $\forall n \in \mathbb{N}^*$, $0 < a < a_n$,

$$b_n = 1 - \frac{a_{n+1}}{a_n} = \frac{a_n - a_{n+1}}{a_n} < \frac{a_n - a_{n+1}}{a}. \text{ 因为}$$

$$\sum_{k=1}^n b_k < \sum_{k=1}^n \frac{a_k - a_{k+1}}{a} = \frac{a_1 - a_{n+1}}{a} < \frac{a_1}{a}, \text{ 所以正项级数 } \sum_{n=1}^{\infty} b_n \text{ 收}$$

敛, 这与已知条件矛盾. 因此, 假设不成立, 即必有 $\lim_{n \rightarrow \infty} a_n = 0$.

3. 解析: 设 M, m 分别是无限实数集 S 的上、下界. 假设 S 没有聚

点, 则 $\forall x \in [m, M]$, 存在邻域 U_x 使得 $U_x \cap S$ 为有限集. 因此,

$\{U_x\}_{x \in [m, M]}$ 构成了闭区间 $[m, M]$ 的一个开覆盖. 由有限覆盖定理可

知, 存在 $[m, M]$ 的有限子覆盖 $\{U_{x_1}, U_{x_2}, \cdots, U_{x_N}\}$, 即

$$S \subset [m, M] \subset \bigcup_{i=1}^N U_{x_i}. \text{ 但是, } \bigcup_{i=1}^N U_{x_i} \text{ 是有限集, 这与 } S \text{ 为无限集矛盾.}$$

因此, 假设不成立, 即 S 有聚点.

4. 解析:

$$\begin{aligned}
& \lim_{x \rightarrow 0^+} \frac{(\sin x)^{\sin x} - x^{\sin x}}{\sin^2 x \arcsin x} \\
&= \lim_{x \rightarrow 0^+} \frac{e^{\sin x \ln(\sin x)} - e^{\sin x \ln x}}{x^3} = \lim_{x \rightarrow 0^+} e^{\sin x \ln x} \frac{e^{\sin x \ln(\frac{\sin x}{x})} - 1}{x^3} \\
&= \lim_{x \rightarrow 0^+} e^{\sin x \ln x} \cdot \lim_{x \rightarrow 0^+} \frac{e^{\sin x \ln(\frac{\sin x}{x})} - 1}{x^3} = e^{\lim_{x \rightarrow 0^+} \sin x \ln x} \cdot \lim_{x \rightarrow 0^+} \frac{\sin x \ln(\frac{\sin x}{x})}{x^3} \\
&= e^{\lim_{x \rightarrow 0^+} x \ln x} \cdot \lim_{x \rightarrow 0^+} \frac{x \ln\left(1 + \frac{\sin x}{x} - 1\right)}{x^3} = e^{\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}} \cdot \lim_{x \rightarrow 0^+} \frac{\sin x - x}{x^3} \\
&= e^{\lim_{x \rightarrow 0^+} -\frac{1}{x^2}} \cdot \lim_{x \rightarrow 0^+} \frac{-\frac{1}{3!}x^3 + o(x^3)}{x^3} = e^{\lim_{x \rightarrow 0^+} (-x)} \cdot \lim_{x \rightarrow 0^+} \frac{-\frac{1}{3!}x^3}{x^3} \\
&= e^0 \cdot \left(-\frac{1}{6}\right) = -\frac{1}{6}
\end{aligned}$$

5. 解析: $\forall \varepsilon > 0$, 因为 $\lim_{x \rightarrow +\infty} f(x)$ 存在, 所以由柯西收敛准则可知, $\exists X > 0$ 使得 $x_1, x_2 > X$ 时, $|f(x_1) - f(x_2)| < \varepsilon$. 又因为 $f(x)$ 在闭区间 $[a, X+1]$ 上连续, 所以 $\exists \delta_1 > 0$ 使得 $x_1, x_2 \in [a, X+1]$ 且 $|x_1 - x_2| < \delta_1$ 时, $|f(x_1) - f(x_2)| < \varepsilon$. 取 $\delta_2 = \min\{\delta_1, 1\}$, 则当 $x_1, x_2 \in [a, +\infty)$ 且 $|x_1 - x_2| < \delta_2$ 时, $|f(x_1) - f(x_2)| < \varepsilon$, 即 $f(x)$ 在 $[a, +\infty)$ 上一致连续.

6. 解析: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\frac{1}{1+t}}{2t+2} = \frac{1}{2(1+t)^2},$

$$\frac{d^2 y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{-1}{(1+t)^3}}{2(1+t)} = -\frac{1}{2(1+t)^4},$$

$$\left. \frac{d^2 y}{dx^2} \right|_{t=0} = -\frac{1}{2}.$$

7. 解析: 设 $F(x) = f(x)(x-b) - f(a)x$, 则

$$F(a) = f(a)(a-b) - f(a)a = -f(a)b,$$

$$F(b) = f(b)(b-b) - f(a)b = -f(a)b, \text{ 因此 } F(a) = F(b). \text{ 由罗尔}$$

中值定理可知, 存在 $\xi \in (a, b)$ 使得 $F'(\xi) = 0$, 即

$$f'(\xi)(\xi-b) + f(\xi) - f(a) = 0, \quad f'(\xi) = \frac{f(\xi) - f(a)}{b - \xi}, \text{ 得证.}$$

8. 解析: 由泰勒中值定理, $\exists \xi_1 \in \left(a, \frac{a+b}{2}\right)$ 和 $\xi_2 \in \left(\frac{a+b}{2}, b\right)$ 使得

$$f\left(\frac{a+b}{2}\right) = f(a) + f'(a)\left(\frac{b-a}{2}\right) + \frac{f''(\xi_1)}{2!}\left(\frac{b-a}{2}\right)^2 = f(a) + \frac{f''(\xi_1)}{2!}\left(\frac{b-a}{2}\right)^2$$

$$f\left(\frac{a+b}{2}\right) = f(b) + f'(b)\left(\frac{a-b}{2}\right) + \frac{f''(\xi_2)}{2!}\left(\frac{a-b}{2}\right)^2 = f(b) + \frac{f''(\xi_2)}{2!}\left(\frac{a-b}{2}\right)^2$$

$$\text{因此, } 0 = f(b) - f(a) + \frac{(b-a)^2}{8} [f''(\xi_2) - f''(\xi_1)],$$

$$\frac{(b-a)^2}{8} [f''(\xi_1) - f''(\xi_2)] = f(b) - f(a),$$

$$\begin{aligned} \frac{4}{(b-a)^2} |f(b) - f(a)| &= \frac{|f''(\xi_1) - f''(\xi_2)|}{2} \leq \frac{|f''(\xi_1)| + |f''(\xi_2)|}{2} \\ &\leq \max\{|f''(\xi_1)|, |f''(\xi_2)|\} \end{aligned}$$

$$\text{因此, } \exists \xi \in (a, b), \text{ 使得 } |f''(\xi)| \geq \frac{4}{(b-a)^2} |f(b) - f(a)|.$$